# MatRROMA User's Manual

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# 1 Introduction

This document introduces the MatRROMA software, developed in MATLAB, which enables the identification of dynamic properties of structures using the System Subspace Identification algorithm. This software facilitates identification through measurements where all study points of the structure have been measured simultaneously, as well as in cases where measurements have been fragmented into multiple measurement sets. Regarding measurements that have been fragmented into measurement sets, the software allows identification if there is at least one measurement point common to all sets (Reference-Based SSI) and also if there is at least one sensor common to two or more sets throughout the entire concatenation of sets (Roving-References SSI) [1].

If you use MatRROMA please cite [1] in your work (books, articles, reports, etc.).

# 2 Requirements

- **Environment:** MATLAB. The software will execute on any operating system supported by MATLAB (Windows, macOS, Linux).
- **Required Toolboxes:** Signal Processing Toolbox.

To run the software, add the source folder to the MATLAB search path (e.g., using the `addpath` command) and execute the routines directly from your working data directory.

# 3 Input variables

The main arguments of the `main` function are the following:

```
matrroma(filenames,q,sampling_rate,setref,method,max_order,varargin)
```

## 3.1 Required variables

These are the variables required to be entered by the user to perform the identification and must be entered in this order:

- `filenames`. Name of the file that lists the .dat or .txt measurement files. This file must also contain the corresponding setup assignment, sensor ID, and the azimuth and elevation angles for each sensor.
- `q`. Maximal system order (`q` $\in \mathbb{N}$). It corresponds to the number of block rows of the Hankel matrices. The selection of this value is not a straightforward decision, it depends on several factors such as the duration of the measurements, the noise level present on them, or the complexity of the structure, among others.
- `sampling_rate`. Sampling rate of the recorded measurements (`sampling_rate` $\in \mathbb{N}$).
- `setref`. Master set used as a baseline to merge all other sets. (`setref` $\in \mathbb{N}$).
- `method`. SSI algorithm that will be used in the identification. The three valid arguments are:
  - **COVC:** Covariance-based SSI algorithm using correlation matrices.
  - **COVD:** Covariance-based SSI algorithm using the subspace matrix.
  - **UPC:** Unweighted Principal Component SSI algorithm.
- `max_order`. Maximal system order that is shown in the stabilization diagram.

## 3.2 Parameters

Unlike the required variables, these arguments are optional, and must be introduced with the name of the parameter to be modified along with its desired value, separated by a comma.

- `l`. This variable correspond to the order noise level of the measurements. By default it is set as 0.
- `refsens4sinset`. Variable used in case of single-setup measurements (synchronous recording of all positions). The definition of reference sensors in single-setup configurations can be used for reducing the size of the Hankel matrices in the identification process. For a single setup configuration, all sensors are taken as references by default, but it is recommended to define specific reference sensors for a more accurate and faster execution of the tool.
- `downsampling_rate`. Desired sampling rate to which the the user wants to downsample the measurement data. By default it is set as the sampling rate.

## 3.3 Pole selection GUI

Once the function has started executing, the user will be requested to interact with a GUI (see Figure 1) to select the stable poles. The following tools are available for the user:

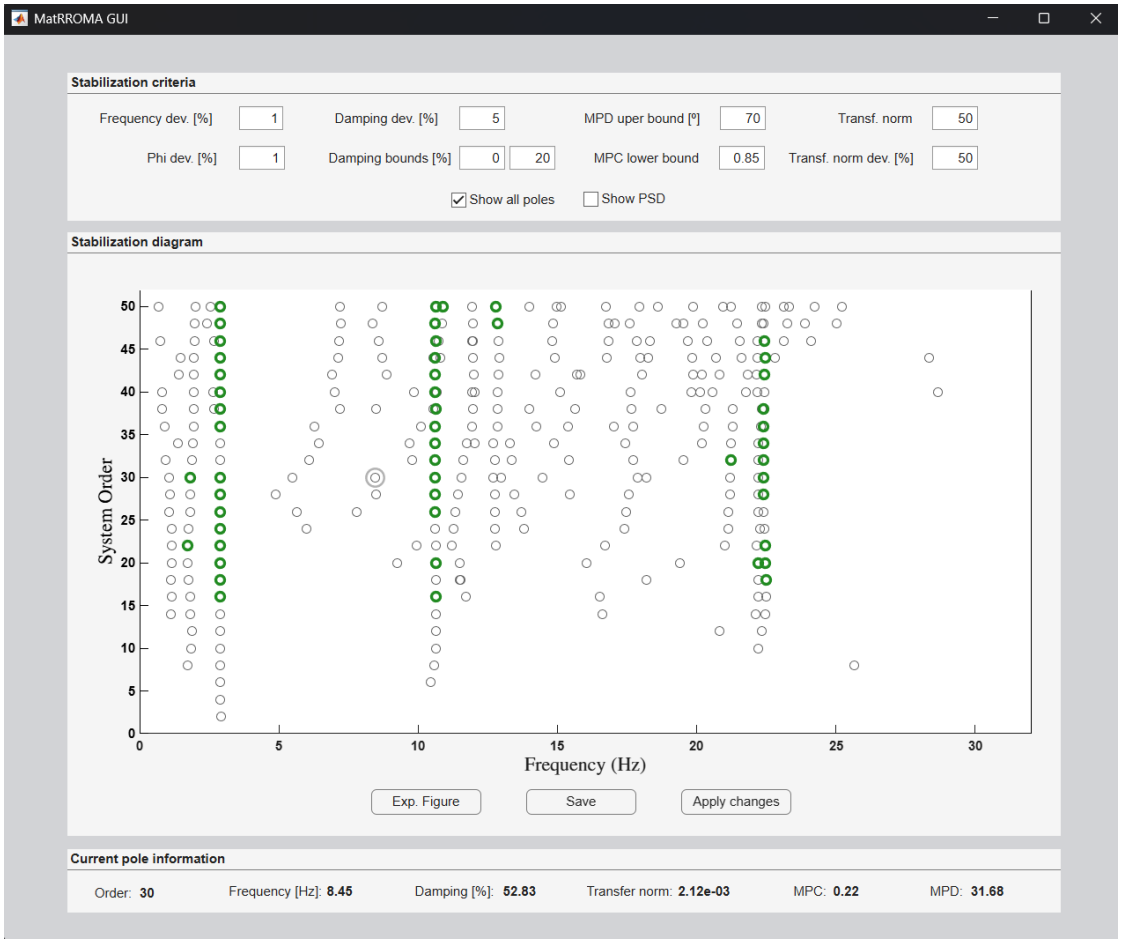



Figure 1: GUI designed for the selection of the desired stable poles

- In the Stabilization criteria panel the user can introduce their desired stabilization criteria, show the first singular values of the Power Spectral Density of the data or show the unstable poles for the introduced parameters by editing and selecting the corresponding items. To apply the selections the user must click the apply button. Available stabilization parameters are:
  - **Frequency Deviation**: Maximum percentage threshold variation in frequency between two consecutive order poles to consider them stable. It is set to 1% by default.

- **Damping deviation**. Maximum percentage threshold variation in modal damping between two consecutive order poles to consider them stable. It is set to 5% by default.
- **Damping limits**. Lower and upper modal damping bound of a pole in which it can be considered stable.
- **Phi deviation**. Maximum percentage threshold variation in mode shapes between two consecutive order poles to consider them stable. It is obtained as 1-MAC*100; where MAC is the Modal Assurance Criterion between two mode shapes. It is set to 1% by default.
- **Modal transfer norm**. Percentage of the Modal Transfer norm of each order of which
- **Modal transfer norm deviation**. Upper cut-off percentage based on the Modal Transfer norm. Only the top percentage of poles with the highest Modal Transfer norm are considered stable. It is set to 50% by default.
- **MPC lower bound**. Minimum Modal Phase Correlation value of a pole to consider it stable. It is set to 0.5 by default.
- **MPD upper bound**. Maximum admissible Modal phase deviation angle (º) to consider a pole stable.

- In the pole information panel, it is possible to view the details of a specific pole by hovering the mouse over it.
- To select or unselect a certain pole, the left mouse button must be clicked over it. Once all the desired poles are selected, the user must press the save button to save the selected poles.
- The stabilization diagram can be saved by clicking on the Exp. fig button.

### 3.4 Input files

- `filenames`: A plain text file detailing the measurement data structure, where each row corresponds to a single sensor. It must follow this specific format:
  - The first row for a given data file must contain 5 columns: the name of the file, its corresponding setup identification (Setup ID), the sensor ID, its azimuth angle, and its elevation angle.
  - For every additional sensor present in that same data file, a new row must be added containing only 3 columns: the sensor ID, azimuth, and elevation angles.

  Once all the sensors of a file are detailed, the sequence starts again on a new row with the name of the next file, its setup ID, and its first sensor details. An example of this row-by-row structure inside the `.txt` file is as follows:

```
data_file_1.txt   1   1   0    0
                      1   90   0
                      2   0    0
data_file_2.txt   2   2   0    90
                      3   90   90
```

- **Measurement data files**: Each file must contain strictly the measured data, where each measurement channel is organized into a separate column.

## 4 Outputs

After executing the function, a .mat variable is obtained with the following information:

1. The `Dyn_par` structure variable with the following parameters:
   - `order`. It shows the system order to which each selected pole belongs.
   - `freq`. Frequency (Hz) at which each selected pole appears.
   - `xi`. Modal damping ratio (%) of each pole.
   - `MPC`. Computed modal phase collinearity (between 0 and 1) of each pole mode shape.

- `MPD`. Modal phase deviation (°) of each pole mode shape.
- `phi`. Mode shape of each pole.

2. The `Point_DOF_order` matrix that shows in each row the sensors IDs and their corresponding remeasurement angles in the same order as the mode shapes.
3. The `ref` variable that indicates the reference sensors used in the identification. For a roving-reference configuration, it always takes a NaN value.

# 5 Auxiliary functions

## 5.1 `matrroma_plotms`

The `matrroma_plotms` function shows an animation of the identified mode shapes. Arguments of the function are the following:

```
matrroma_plotms(MSname,coordsname,conectname)
```

- `MSname`. Name of the `.mat` file saved at the end of the `Main` function execution.
- `coordsname`: A plain `.txt` file containing as many rows as sensor positions in the global assembly, and four columns with the following information (delimited by spaces):

  SensorID Xcoordinate Ycoordinate Zcoordinate

- `conectname`: A plain `.txt` file that defines the geometry. It contains as many rows as connections between sensors, and either two or three columns depending on the desired type of representation:
  - **Wireframe**: Defined by two columns of sensor IDs. Each row represents a single line segment connecting two sensors. It is necessary to define all the line segments in the structure.
  - **Solid/Mesh**: Defined by three columns of sensor IDs. The mesh is represented by triangles. It is mandatory that the three nodes of each triangle are always defined following the same rotational direction.

## 5.2 `matrroma_poser`

The `matrroma_poser` function merges the identified modes from several individual setups in order to build a global identification. The only argument of this function is a string vector containing the names of the `.mat` files saved at the end of the `main` function execution for each setup. For a correct execution, it is necessary that all setups share the same reference sensor. An example of a call to this function is:

```
matrroma_poser(["idpar1.mat","idpar2.mat"...])
```

# 6 Operation example

To demonstrate the application of the code, this section presents the results obtained from the experimental data of an elevated pedestrian crossing over a highway, previously introduced in [1]. All files required to reproduce this example are provided in the "Pedestrian Crossing Over a Highway Example" folder of the GitHub repository, while the dataset is also available via Zenodo [2].
In this analysis, the maximum system order is set to 50, the COVD method is applied, and $S_1$ serves as the master set. Stable physical modes are identified using the default stabilization criteria: a maximum tolerance of 1% for natural frequency, 5% for damping ratio, and 1% for the MAC value. Finally, the representative pole for each mode is chosen based on the lowest MPD and the highest MPC. Figure 1 presents the GUI designed for defining the stabilization criteria and selecting the desired stable poles.

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## References

[1] Samuel González-Jiménez, Luis A. Padrón, Guillermo M. Álamo, and Jacob D. R. Bordón. Operational modal analysis from multi-setup measurements with roving references and stochastic subspace identification. *Engineering Structures*, 367:123529, 2026.

[2] Samuel González-Jiménez, Luis A. Padrón, Guillermo M. Álamo, and Jacob D. R. Bordón. Dataset employed in "operational modal analysis from multi-setup measurements with roving references and stochastic subspace identification". Zenodo, lul 2026. Url: https://doi.org/10.5281/zenodo.15545604.